Emojis & the metalinguistic performance of LLMs

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1 Introduction

ChatGPT is very good at giving acceptability judgments for many English sentences (for some preliminary studies, see Collins 2024a; b). In some ways, ChatGPT's language use very closely resembles that of humans (Cai et al. 2023; Ortega-Martín et al. 2023; J. Wang et al. 2023); however, there are many ways in which ChatGPT and other LLMs like it fail to linguistically perform on the same level as humans (Borji 2023; Basmov & Goldberg & Tsarfaty 2023; Jang & Lukasiewicz 2023; Shen et al. 2023; Zhong et al. 2023; Zuccon & Koopman 2023, a.o.). These findings suggest that, while ChatGPT and other LLMs are able to generate human-like linguistic utterances, they do not have an internal hierarchical syntax (Linzen & Baroni 2021; Contreras Kallens et al. 2023; H. Zhou et al. 2023; Hale & Stanojević 2024; Manova 2024, a.o).

Emojis are pictorial elements that exist in the digital keyboard layout of most smartphones and computers. They are easily integrated with text and, as such, appear often in written language in computer-mediated communication (CMC). Emojis often appear in written utterances as morphosyntactic constituents (Storment 2024), a phenomenon known as *pro-text emojis*. There is strong evidence to suggest that pro-text emojis are fully integrated into linguistic systems in terms of their morphosyntax rules (Storment 2024) and their mechanisms of semantic interpretation (Cohn et al. 2018; Tieu et al. 2023). Emojis are unique as an incredibly widespread, conventionalized set of symbols and pictograms that can be readily incorporated into written natural language. As such, they offer an incredibly meaningful glimpse at the way human language interfaces with the cognitive systems that allow for the interpretation of iconic and pictorial elements, and visual information in general.

In this short paper I present novel data from GPT-4o (OpenAI 2024) handling sentences containing pro-text emojis in English and Spanish. I show that, while GPT-4o is able to accurately identify what many – but crucially not all – emojis mean, it is completely unable to predict with any accuracy where pro-text emojis can acceptably appear in written utterances.

Human cognition allows for the extraction of meaning from visual stimuli, and this seems to be strongly connected to the human language faculty when we look at the existence of language phenomena such as iconicity, partial iconicity, pictorial symbols in orthography, gesture, and, of course, the existence of signed languages. ChatGPT is crucially missing that aspect of the human capacity for language, and it seems to have no functional alternative, even for visual elements that appear in text. This gap – along with the gaps in syntactic awareness – spells doom for LLMs when it comes to the treatment of visual elements as syntactic constituents. This is a gap that must be filled if generative AI models are to truly match human language performance.

2 Pro-text emojis

Pro-text emojis (borrowing terminology from the semantics of gestures in spoken language (Schlenker 2018)) are emojis that appear inside of an utterance as morphosyntactic constituents. See the following examples from Storment (2024), ultimately from Twitter/X.

(1) a. I need to 🛌 before I see the end of this game or I'll be 😢 I missed it
b. 🏡 is where the ❤️ is
c. Some 🌈 people were discriminated against at protest grounds

Some work on the semantics of pro-text emojis suggest that they replace written words in an utterance (Tieu et al. 2023), but this conclusion is puzzling when considering that pro-text emojis may appear as elements smaller than a completely formed word in a given derivation.

(2) a. He ❤️s to 📚
b. My therapist 👻ed me so I took selfies in the parking lot
c. likeeee the secondhand embarrassment is ☠️ing me
d. A couple of 🚬s smoking 🚬s
e. ⬆️er than the price of gas

It is not the case, though, that pro-text emojis may freely replace any morpheme. Storment (2024) systematically demonstrates that there are licit and illicit morphosyntactic positions in which pro-text emojis are licensed, and language users have clear grammaticality judgments about what these positions are.

(3) a. *I like 🌸al perfumes (int: *floral*)
b. *Professor Rambow is a 💻ational linguist (int: *computational*)
c. *🤨ity killed the cat (int: *curiosity*)
d. *I need to 💪en my core (int: *strengthen*)

The examples in (3) are ungrammatical because such forms are unattested online and because English-speaking emoji users consistently judge these forms to be ungrammatical.

There are also restrictions on pro-text emoji placement in other languages, though the restrictions vary from language to language. Take the example of Spanish.

(4) a. Yo te ❤️(*-o) mucho
I you ❤️(*-1SG.PRES) much
'I love you very much'
b. Tú me ❤️(*-s) mucho
you me ❤️(*-2SG.PRES) much
'You love me very much'

While verbal agreement and tense suffixes are possible with pro-text emojis in English (2a-c), they are generally not possible in Spanish. This shows that the combinatorics of pro-text emojis with other morphemes are sensitive to the morphosyntactic structure of the language in which they are embedded. See Storment (2024) for a detailed analysis of this difference between English and Spanish.

These data clearly show two very important facts. First, pro-text emojis do not replace orthographic words. They appear as units smaller than the word-level, yet it is not the case that they can freely stand in for any morpheme. As such, they take part in morphosyntactic operations and must be obedient to the grammatical constraints of the language in which they appear, which is the second important fact.

3 GPT-4o tests

I used OpenAI's ChatGPT interface to perform a series of linguistic experiments with GPT-4o on sentences containing pro-text emojis. I performed these experiments on two languages: English and Spanish. In these experiments, I tested GPT-4o's knowledge on emoji recognition in isolation, emoji recognition embedded in sentences, and metalinguistic acceptability judgments on sentences containing pro-text emojis. I detail the findings of these short experiments here.

3.1 Emoji recognition

GPT-4o is, for the most part, exceptionally good at recognizing emojis. This is consistent with prior research (Y. Zhou et al. 2024). It can identify what an emoji depicts, how it is used, and even what an emoji's Unicode code sequence is. I asked GPT-4o to identify what a given emoji depicts, what its keywords are, and what its Unicode entry is for a random sample of 40 emojis, five from each category of emoji: Smileys and People, Animals & Nature, Food & Drink, Activity, Travel & Places, Objects, Symbols, and Flags. I ran this prompt (shown in (5), followed by a numbered list of forty emojis) three times.

(5) *Emoji recognition prompt*
Identify the meaning, keywords, and Unicode code associated with the following emojis.

GPT-4o was able to identify all information correctly for all emojis in all three tests. In other words, it performed perfectly.

(6) *Emoji recognition test results*

Category	% Correct (Round 1)	% Correct (Round 2)	% Correct (Round 3)
Smileys & People			
Animals & Nature			
Food & Drink			
Activity	100	100	100
Travel & Places			
Objects			
Symbols			
Flags			

However, during some preliminary testing in which I gave GPT-4o the prompt in (7), some of the results were quite different.

(7) *Preliminary testing for emoji recognition*
What is the metadata for the emoji ___? I mean the Unicode entry and keywords that are associated with each emoji.

I did this for thirteen random emojis, and it did not perform perfectly. It failed to identify three emojis: 🦷(tooth), 👂(ear), and 🦵(leg). I show GPT-4o's response to (7) for each of these here.

(8) *Identification of the 🦷 (tooth) emoji* (response from GPT-4o)

The 🦷 emoji represents a **pigeon** and its metadata includes the following:

Unicode:

- **Unicode code point**: **U+1F9F7**

Keywords:

- **Pigeon**
- **Bird**
- **Flying**
- **Urban**
- **Messenger**
- **Animal**
- **Feathers**

The 🦷 emoji was added to Unicode in **2019** as part of Unicode 12.0. It is often used to represent the pigeon itself, or more metaphorically as a symbol of communication, urban life, or even a humorous or casual reference to birds in general.

In addition to misidentifying the emoji, the Unicode code it gave here is for the safety pin (📎) emoji. At the time of writing this, there currently is no pigeon emoji.

(9) *Identification of the 👂 (ear) emoji* (response from GPT-4o)

The 👂 emoji represents a **mouth** and is often used to depict the mouth itself in various contexts. Here is the metadata for this emoji:

Unicode:

- **Unicode code point**: **U+1F442**

Keywords:

- **Mouth**
- **Face**
- **Talking**
- **Speak**
- **Speech**
- **Expression**
- **Lips**

The 👂 emoji was added to Unicode in **2010** as part of Unicode 6.0. It is commonly used to refer to the **mouth** as part of the face, and it can be used in contexts relating to **speaking**, **talking**, or **expressing oneself**. This emoji may also be used to represent **communication**, **expression**, or sometimes in discussions about **physical features** like the lips or speech.

The Unicode code for the 👂 emoji given here is correct, despite misidentifying the meaning.

(10) *Identification of the 🦵 (leg) emoji* (response from GPT-4o)

The 🦵 emoji represents a **monkey face**, specifically a type of monkey, and is typically used in contexts related to monkeys or playful behavior. Here is the metadata for this emoji:

Unicode:

- **Unicode code point**: **U+1F992**

Keywords:

- **Monkey**
- **Monkey face**
- **Animal**
- **Primates**
- **Mammal**
- **Playful**
- **Funny**
- **Cute**
- **Wildlife**

The 🦵 emoji was added to Unicode in **2019** as part of Unicode 12.0. It is used to represent **monkeys** in a playful or humorous way. The emoji may be used in conversations about **animals**, **wildlife**, or **primates** and is often associated with **fun** and **playfulness** due to the typical behavior of monkeys. It can also convey a **cute** or **silly** expression, often in a lighthearted or casual context.

While there is a monkey face emoji that exists, the Unicode code given in (10) is the one for the giraffe (🦒) emoji.

GPT-4o is mostly able to correctly identify emojis, but it still makes mistakes. The mistakes seem inconsistent and unpredictable, though I was able to force it to misidentify these three emojis several more times outside of the context of this formal experiment. Interestingly, the three emojis that I found GPT-4o to struggle with all depict a part of the human body.

3.2 Translation

For the next task, I had GPT-4o "translate" sentences containing pro-text emojis into sentences that only contain standard orthographic words. Using both grammatical and ungrammatical forms, I gave GPT-4o the following prompt along with 40 sentences containing pro-text emojis.

(11) *Translation prompt*

Paraphrase each of the following sentences without using emojis.

In the prompt, I did not indicate which sentences were grammatical and ungrammatical. GPT-4o executed this task perfectly, and accurately conveyed the meaning – or indented meaning – of each sentence.

Here I show a few examples of grammatical and ungrammatical utterances that GPT-4o successfully translated. I indicate ungrammatical sentences here with an asterisk.

(12) a. Prompt: I ❤️ you
GPT-4o: I love you
b. Prompt: She 💋ed me
GPT-4o: She kissed me
c. Prompt: He ❤️s 📚ing
GPT-4o: He loves reading
d. Prompt: I'm going ✈️ the store
GPT-4o: I'm going to the store
e. Prompt: This is getting 👶ish
GPT-4o: This is getting childish

(13) a. Prompt: *I like 🌸al perfumes
GPT-4o: I like floral perfumes
b. Prompt: *I need to 💪en my core
GPT-4o: I need to strengthen my core
c. Prompt: *My son is learning ➕ition in school
GPT-4o: My son is learning addition in school
d. Prompt: *Mr. Kamano is 🇯🇵ese
GPT-4o: Mr. Kamano is Japanese
e. Prompt: *She looks 👼ic but she's actually 😡
GPT-4o: She looks angelic but she's actually angry

This study demonstrates that GPT-4o has an excellent grasp on the semantic content of emojis and their conventions of use, though there is no indication here that it has any notion of syntax or at least morphological combinatorics. I confirm this in the following experiment.

3.3 Acceptability judgments

I then had GPT-4o rate grammatical and ungrammatical utterances for their acceptability. I did this for 50 English sentences and 35 Spanish sentences containing pro-text emojis. I ran each prompt three times. I found that GPT-4o is inaccurate and inconsistent when it comes to giving grammaticality judgments of sentences containing pro-text emojis, and these judgments do not match those from human native speakers.

I gave GPT-4o the following prompt(s). I had to specify "informal {English/Spanish}" because otherwise it judged almost every sentence to be unacceptable.

(14) *Acceptability judgment prompt*
For each sentence, tell me if it is acceptable in informal {English/Spanish} or not. Do not give any explanations.

Preliminary data from Collins (2024a; b) show that GPT-4o is quite good at giving native-speaker-like grammaticality judgments for sentences that do not contain pro-text emojis, and my data here show that GPT-4o's judgments are quite inaccurate for emoji sentences.

I present the full set of tabulated results here. I include a numbered list of sentences (the ungrammatical ones indicated here with an asterisk, which I did not incidate in the prompt I gave to GPT-4o), the acceptability judgments that GPT-4o gave each time, and then a match score, which shows the percentage of correct responses that GPT-4o gave for a given utterance.

(15) *Acceptability judgment responses for English*

Sentences	GPT-4o rating 1	GPT-4o rating 2	GPT-4o rating 3	Match score
1. I ❤️ you	Acceptable	Acceptable	Acceptable	100
2. You ❤️ me	Acceptable	Acceptable	Acceptable	100
3. He ❤️ me	Acceptable	Acceptable	Acceptable	100
4. He ❤️s me	Acceptable	Acceptable	Acceptable	100
5. I ❤️d you	Acceptable	Acceptable	Acceptable	100
6. I 👀 you	Acceptable	Acceptable	Acceptable	100
7. He 👀s me	Acceptable	Acceptable	Acceptable	100
8. He 👀d me	Acceptable	Acceptable	Not acceptable	66.6667
9. He 👀ed me	Not acceptable	Not acceptable	Not acceptable	0
10. She 💋es me	Not acceptable	Not acceptable	Acceptable	33.3333
11. She 💋s me	Acceptable	Acceptable	Acceptable	100
12. She 💋ed me	Acceptable	Acceptable	Acceptable	100
13. She 💋d me	Acceptable	Acceptable	Acceptable	100
14. I love 🥭s	Acceptable	Acceptable	Acceptable	100
15. I love 🥭es	Not acceptable	Not acceptable	Not acceptable	0
16. *Let's visit the 🍷ery	Acceptable	Acceptable	Acceptable	0
17. He is the 🏳️‍🌈est person ever	Acceptable	Acceptable	Acceptable	100
18. *I like 🌸al perfumes	Acceptable	Acceptable	Acceptable	0
19. *I am ready for 🌙ar new year	Acceptable	Acceptable	Acceptable	0
20. Hand me the ✂️s	Acceptable	Acceptable	Acceptable	100
21. Hand me the ✂️	Acceptable	Not acceptable	Acceptable	66.6667
22. He ❤️s to 📚	Acceptable	Acceptable	Acceptable	100
23. He ❤️s 📚ing	Acceptable	Acceptable	Acceptable	100
24. *I need to 💪en my core	Acceptable	Not acceptable	Acceptable	33.3333
25. He's been a 🚬er for 20 years	Acceptable	Acceptable	Acceptable	100
26. This makes me want to 😭 my eyes out	Acceptable	Acceptable	Acceptable	100
27. He's such an 🍑💋er	Acceptable	Not acceptable	Acceptable	66.6667
28. I'm going ✈️ the store	Not acceptable	Not acceptable	Not acceptable	0
29. *He studies 💻ational linguistics	Acceptable	Acceptable	Acceptable	0
30. *We are 💈ar opposites	Acceptable	Acceptable	Acceptable	0
31. My therapist 👻ed me	Acceptable	Acceptable	Acceptable	100
32. My therapist 👻d me	Not acceptable	Acceptable	Acceptable	66.6667
33. He is 🥺ing at me	Acceptable	Acceptable	Acceptable	100
34. She 🥺ed at me	Acceptable	Acceptable	Acceptable	100
35. *I am 🇺🇸n	Acceptable	Acceptable	Acceptable	0

36. This is getting 👶ish	Acceptable	Acceptable	Acceptable	100
37. *She looks 😇ic but she's actually 😡	Acceptable	Acceptable	Acceptable	0
38. *My son is learning ➕ition in school	Not acceptable	Acceptable	Not acceptable	66.6667
39. *🤨ity killed the cat	Not acceptable	Not acceptable	Not acceptable	100
40. *Mr. Kamano is 🇯🇵ese	Not acceptable	Acceptable	Not acceptable	66.6667
41. Wow, 🫵 must really love 🫵self	Not acceptable	Not acceptable	Acceptable	33.3333
42. Wow, 🫵 must really love yourself	Acceptable	Acceptable	Acceptable	100
43. *Wow, you must really love 🫵self	Not acceptable	Not acceptable	Acceptable	33.3333
44. I think 🫵 left 🫵 phone at the bar	Not acceptable	Not acceptable	Acceptable	33.3333
45. I think 🫵 left your phone at the bar	Acceptable	Acceptable	Acceptable	100
46. *I think you left 🫵 phone at the bar	Not acceptable	Not acceptable	Acceptable	66.6667
47. I love 🫵 very much	Not acceptable	Not acceptable	Acceptable	33.3333
48. Do 🫵 love me?	Not acceptable	Not acceptable	Acceptable	33.3333
49. I gave 🫵 the book yesterday	Not acceptable	Not acceptable	Acceptable	33.3333
50. I gave the book to 🫵 yesterday	Not acceptable	Not acceptable	Acceptable	33.3333

The average match score for all English sentences is 63.3332.

(16) *Acceptability judgment responses for Spanish*

Sentences	GPT-4o rating 1	GPT-4o rating 2	GPT-4o rating 3	Match score
1. Te ❤️	Acceptable	Acceptable	Acceptable	100
2. *Te ❤️o	Not Acceptable	Acceptable	Acceptable	0
3. *Te ❤️o mucho	Acceptable	Acceptable	Acceptable	0
4. Te ❤️ mucho	Acceptable	Acceptable	Acceptable	100
5. Yo te ❤️ mucho	Acceptable	Acceptable	Acceptable	100
6. *Yo te ❤️o mucho	Not Acceptable	Acceptable	Acceptable	33.3333
7. Me ❤️	Not Acceptable	Acceptable	Not acceptable	33.3333
8. *Me ❤️s mucho	Acceptable	Acceptable	Acceptable	0
9. *Me ❤️as mucho	Not Acceptable	Acceptable	Not acceptable	66.6667
10. *Tú me ❤️s mucho	Acceptable	Acceptable	Acceptable	0
11. Nosotros te ❤️ mucho	Acceptable	Acceptable	Acceptable	100
12. Nosotros te ❤️mos mucho	Not Acceptable	Acceptable	Acceptable	66.6667
13. *Nosotros te ❤️amos mucho	Acceptable	Not acceptable	Not acceptable	66.6667
14. Te ❤️mos mucho	Not Acceptable	Acceptable	Acceptable	66.6667
15. *Te ❤️amos mucho	Acceptable	Acceptable	Acceptable	0
16. Aquí están los 🐶itos	Acceptable	Acceptable	Acceptable	100
17. Aquí están los 🐶	Acceptable	Acceptable	Acceptable	100
18. Aquí están los 🐶s	Not Acceptable	Acceptable	Acceptable	66.6667
19. *Aquí están los 🐶os	Not Acceptable	Not acceptable	Not acceptable	100
20. Las 🍎 son rojas	Acceptable	Acceptable	Acceptable	100
21. Las 🍎s son rojas	Not Acceptable	Acceptable	Acceptable	66.6667
22. *Las 🍎as son rojas	Not Acceptable	Not acceptable	Not acceptable	100

23. Yo vi el ✈️azo	Not Acceptable	Acceptable	Acceptable	66.6667
24. Yo vi el ☀️azo	Not Acceptable	Acceptable	Acceptable	66.6667
25. Yo vi el ☀️ito	Acceptable	Acceptable	Acceptable	100
26. Yo vi el ☀️cito	Not acceptable	Acceptable	Acceptable	66.6667
27. Aquí está el 🐶	Acceptable	Acceptable	Acceptable	100
28. *Aquí está el 🐶o	Not acceptable	Not acceptable	Acceptable	66.6667
29. Aquí está la 🐶	Acceptable	Acceptable	Acceptable	100
30. *Aquí está la 🐶a	Not acceptable	Not acceptable	Not acceptable	100
31. Aquí están las 🐶s	Acceptable	Acceptable	Acceptable	100
32. *Aquí están las 🐶as	Not acceptable	Not acceptable	Not acceptable	100
33. Quiero tomar un ☕cito	Acceptable	Acceptable	Acceptable	100
34. Quiero tomar un ☕ito	Not acceptable	Acceptable	Acceptable	66.6667
35. *Yo te ❤️é antes	Not acceptable	Acceptable	Acceptable	33.3333

The average match score for all Spanish sentences is 69.5238.

GPT-4o performed slightly better with the Spanish examples than it did with the English examples. Interestingly, this is parallel with the observation in Storment (2024) that native Spanish speakers have more robust judgments on sentences containing pro-text emojis in Spanish than do English speakers in English. More research and experimentation is necessary to confirm if there is actual correlation there, or if it is just coincidence.

Overall, GPT-4o gave inaccurate ratings for these sentences. It especially struggled to label the ungrammatical examples as such, though it also inaccurately labeled many grammatical sentences. In the following section I discuss the theoretical implications of this both for human language and for generative AI systems.

4 Theoretical implications

Emoji combinatorics – that is, where an emoji may appear within a given utterance – reveal an syntax that these emojis must abide by. There is some internal hierarchical structure in each language that licenses these visual elements in some locations, but not others. The distribution of these elements forces us to consider the syntactic structures of natural language.

LLMs, however, operate over strings of text; there is no evidence that have an internal hierarchical syntax, and, as such, do not operate on any hierarchically-ordered syntactic constituents (Manova 2024). This raises two important questions. First, what parameters does the LLM use to determine the grammaticality of these utterances containing pro-text emojis? Second, how does it determine the grammaticality of any utterance? Surely these systems use the same metrics to determine the grammaticality of the emoji utterances than they would any other.

LLMs like GPT-4o determine the grammaticality of utterances by generalizing over the data in their training sets. If a given form frequently appears in the system's training data, it is more likely to accept such forms as grammatical. Conversely, if a given form is poorly attested or completely unattested in the training data, it is less likely to accept those forms.

Frequency and attestation is one of the metrics used in Storment (2024) to determine the acceptability of a given pro-text emoji form. In that paper, grammatical forms are well-attested online (in addition to being judged grammatical by native speakers), and ungrammatical forms are either poorly attested or not attested (also in addition to being judged ungrammatical by native speakers). As shown in the numerous studies cited in the introduction of this paper, the training data works particularly well in many cases to cause the LLM to come to the correct conclusions on grammaticality, though it is not perfect.

In fact, the metrics of attestation and frequency are quite accurate in Storment (2024) for determining the grammaticality of emoji sentences, so one might expect a greater degree of accuracy from the LLMs in this way. This forces us to wonder why GPT-4o struggles particularly with these data containing pro-text emojis, assuming it contains such utterances in its training set. If it does not, then this is an obvious way in which the model can improve, though I assume it does have some exposure to this kind of data because it can interpret the sentences successfully and because the model is trained on 175 billion parameters.

Generative AI systems do not have the capacity for processing iconic and pictorial elements, at least not directly. Obviously, they do not have the means of directly perceiving and processing visual and auditory stimuli in the way that that humans do. Whatever iconicity is (Davidson 2023), and however the human brain processes that information, LLMs lack that same resource.

Human language readily integrates iconic and pictorial elements, be it depictive sounds (Wiese 1996), gestures (Goldin-Meadow & Brentari 2017; Schlenker 2018), emojis (Grosz & Kaiser & Pierini 2021; Storment 2024), or even entire pictures: see the following examples of Internet memes containing entire pictures as syntactic constituents.

(17) a.

b.

These examples show pictures as DPs (*roses*), NPs (*elephant, Shire*), AdjPs (*red*), and CPs (*The ability to speak does not make you intelligent, Harry, did you put your name in the Goblet of Fire?*). Interpreting these memes relies not only on the depictive iconicity of the images, but also on the reader's knowledge of *Star Wars* (17a) and the *Lord of the Rings* and *Harry Potter* movies (17b). Preliminary data suggests that GPT-4o struggles to interpret these memes as well.

Storment (2024) shows that the way in which these elements are incorporated into language is indicative of some underlying internal structure (this idea is expressed elsewhere, such as Wiese (1996)). The ability to incorporate iconic visual elements into language perhaps relies on the ability of these elements to be embedded in a syntax. There is a generalization here that these generative AI systems do not have syntactic awareness, and that is why GPT-4o is unable to make generalizations about the combinatorics of pro-text emojis.

5 Conclusion

The use of emojis and other pictorial elements is an informatic tool, a tool that forces us to consider the morphosyntactic constraints of a given language. We can use this tool to see where the limitations of LLMs' comprehension lie.

In this paper I show that the LLM GPT-4o cannot make accurate generalizations about the grammaticality of visual elements that are embedded inside utterances in English and Spanish, even though it, for the most part, can very accurately interpret the semantic content of emojis and the utterances in which they appear.

This paper is meant to lay the foundations for future research on this subject. The data and theoretical discussion presented here are still very preliminary, and they introduce many interesting questions concerning iconicity in syntax, human cognition, and the improvement of generative AI.

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